Ex 3 — Data Visualization

Dataset: University Admissions (admissions.csv)
Reg No: URK24CS1021

```
print("URK24CS1021")

import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_theme(style="whitegrid")
plt.rcParams["figure.figsize"] = (7, 4.5)

df = pd.read_csv("admissions.csv")
df
```

URK24CS1021

	Student_ID	GPA	SAT_Score	ACT_Score	Extracurriculars	Recommendation_Strength	Admitted
0	U001	3.8	1420	32	3	Strong	Yes
1	U002	3.2	1200	26	1	Weak	No
2	U003	4.0	1560	35	5	Strong	Yes
3	U004	3.5	1310	29	2	Moderate	Yes
4	U005	3.9	1480	33	4	Strong	Yes
5	U006	3.1	1150	24	1	Weak	No
6	U007	3.7	1380	30	3	Moderate	Yes
7	U008	3.0	1100	22	0	Weak	No
8	U009	4.0	1520	34	5	Strong	Yes
9	U010	3.6	1350	28	2	Moderate	Yes
10	U011	3.3	1250	27	1	Weak	No
11	U012	3.8	1450	31	4	Strong	Yes
12	U013	3.4	1280	27	2	Moderate	No
13	U014	3.9	1500	33	5	Strong	Yes
14	U015	3.2	1180	25	1	Weak	No

```
print("URK24CS1021")

df.dtypes
```

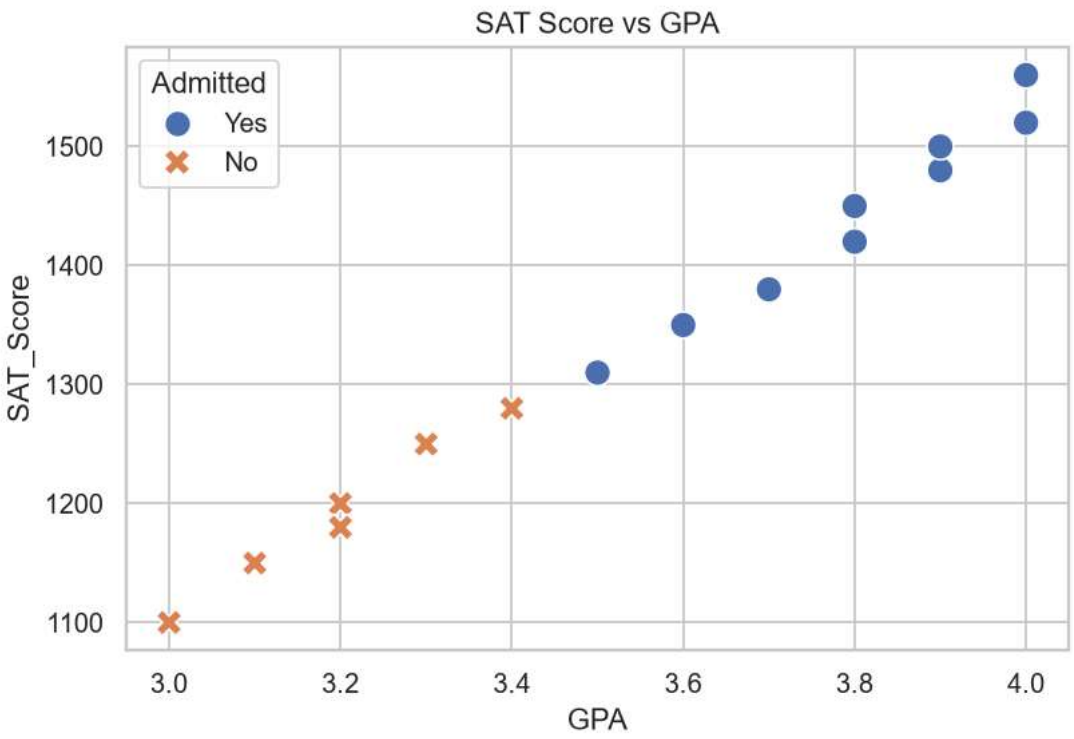
```
URK24CS1021
Student_ID                object
GPA                      float64
SAT_Score                  int64
ACT_Score                  int64
Extracurriculars           int64
Recommendation_Strength    object
Admitted                   object
dtype: object
```

1. Scatter plot — does a higher GPA come with a higher SAT score?

```
print("URK24CS1021")

sns.scatterplot(data=df, x="GPA", y="SAT_Score", hue="Admitted", style="Admitted", s=120)
plt.title("SAT Score vs GPA")
plt.show()
```

```
URK24CS1021
```

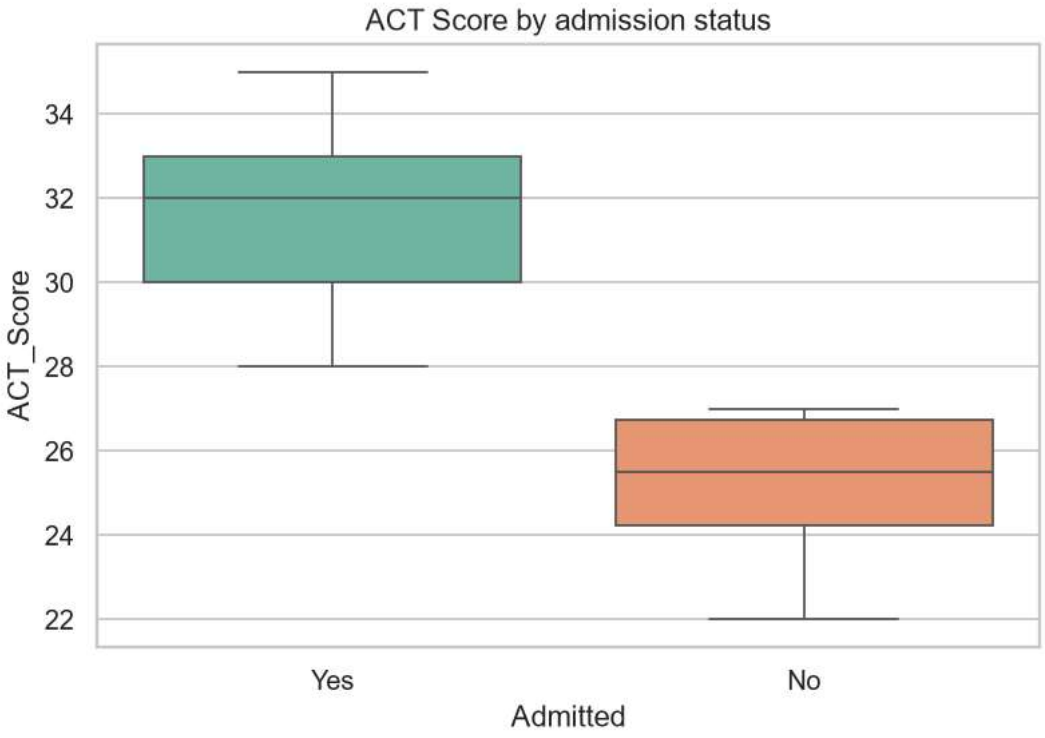


2. Box plot — how do ACT scores differ between admitted and rejected?

```
print("URK24CS1021")

sns.boxplot(data=df, x="Admitted", y="ACT_Score", hue="Admitted", palette="Set2", legend=False)
plt.title("ACT Score by admission status")
plt.show()
```

```
URK24CS1021
```



3. Bar plot — does a stronger recommendation mean a better chance?

```
print("URK24CS1021")

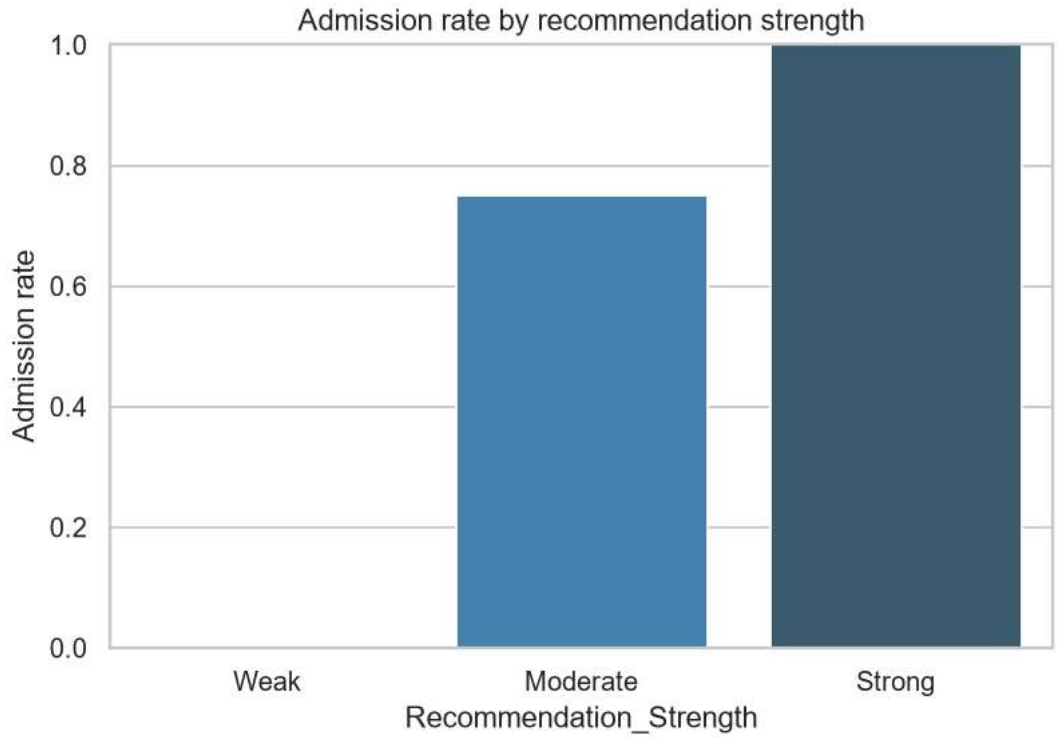
rate = (df.groupby("Recommendation_Strength")["Admitted"]
          .apply(lambda s: (s == "Yes").mean())
          .reindex(["Weak", "Moderate", "Strong"]))
rate
```

```
URK24CS1021
Recommendation_Strength
Weak        0.00
Moderate    0.75
Strong      1.00
Name: Admitted, dtype: float64
```

```
print("URK24CS1021")

sns.barplot(x=rate.index, y=rate.values, hue=rate.index, palette="Blues_d", legend=False)
plt.ylabel("Admission rate")
plt.ylim(0, 1)
plt.title("Admission rate by recommendation strength")
plt.show()
```

```
URK24CS1021
```

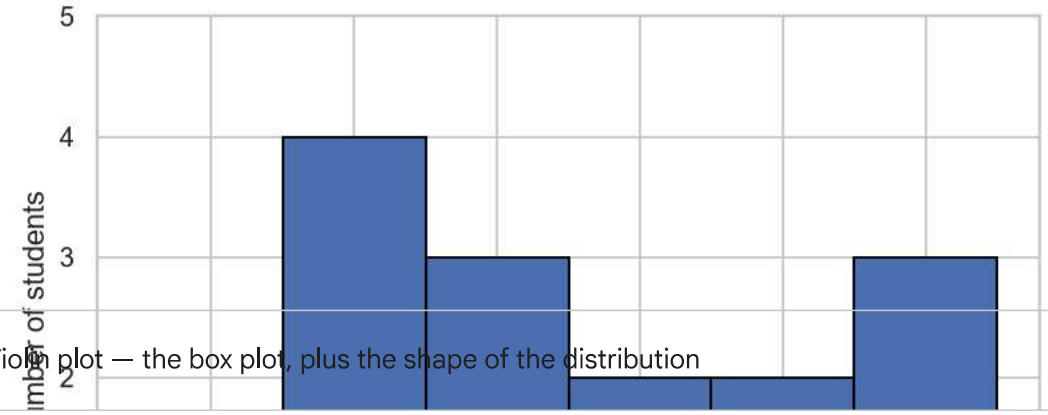


4. Histogram — how many activities does a typical applicant have?

```
print("URK24CS1021")

plt.hist(df["Extracurriculars"], bins=range(0, 7), edgecolor="black", align="left")
plt.xticks(range(0, 6))
plt.yticks(range(0, 6))
plt.xlabel("Number of extracurriculars")
plt.ylabel("Number of students")
plt.show()
```

```
URK24CS1021
```



5. Violin plot — the box plot, plus the shape of the distribution

```
print("URK24CS1021")

sns.violinplot(data=df, x="Admitted", y="GPA", hue="Admitted", palette="Set3", legend=False)
plt.title("GPA distribution by admission status")
plt.show()
```

```
URK24CS1021
```

